

Section 11.1: Sequences

In the course Math-1720, we discussed the limit $\lim_{x \rightarrow \infty} f(x) = L$. The equality means that when x gets larger and larger, the value $f(x)$ gets closer and closer to L . We also defined $\lim_{x \rightarrow \infty} f(x) = \pm\infty$.

Now we turn to infinite sequences of real numbers and consider limits of sequences. E.g., the following are sequences of real numbers:

$$1, 3, 5, 7, \dots; \quad 2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \dots; \quad \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \frac{1}{2^4}, \dots$$

For a general sequence, we write it as $\{a_n\}_{n=1}^{\infty}$, or $\{a_n\}$, or

$$a_1, a_2, a_3, a_4, \dots$$

So, we can write the above three sequences as $a_n = 2n - 1$, $b_n = \frac{n+1}{n}$, $c_n = \frac{1}{2^n}$.

In fact, a sequence is a function with domain $\mathbb{N}$ (the set of positive integers). For such function $s : \mathbb{N} \rightarrow (-\infty, \infty)$, we often write $a_n = s(n)$ and express the function s as the sequence $\{a_n\}$. Then we can consider the graph of a sequence. And we can consider the limit of a sequence: if L is a real number, then

$\lim_{n \rightarrow \infty} a_n = L \text{ means as } n \text{ gets larger and larger, } a_n \text{ gets closer and closer to } L.$

We also write $a_n \rightarrow L$ when $\lim_{n \rightarrow \infty} a_n = L$. In this case, we say that the sequence $\{a_n\}$ is convergent with limit L . We say that the sequence $\{a_n\}$ is divergent if $\{a_n\}$ is not convergent.

Similarly, we can define $\lim_{n \rightarrow \infty} a_n = \pm\infty$. Note that in this case, $\{a_n\}$ is divergent.

Also, we have the ε - N description of $\lim_{n \rightarrow \infty} a_n = L$ and its interpretation by the graph of the sequence $\{a_n\}$.

For the above three sequences $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$, we have

$$a_n \rightarrow \infty \text{ and hence } \{a_n\} \text{ is divergent,} \quad b_n \rightarrow 1, \quad c_n \rightarrow 0.$$

Relations between Functions and Sequences.

- (i) If $f : [1, \infty) \rightarrow (-\infty, \infty)$, then $a_n = f(n)$ defines a sequence $\{a_n\}$.
- (ii) If $\{a_n\}$ is a sequence, then there is $f : [1, \infty) \rightarrow (-\infty, \infty)$ satisfying $f(n) = a_n$.

Note. The function f in (ii) is not unique.

Theorem. Let f be a function and $a_n = f(n)$.

- (1) If $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} a_n = L$.
- (2) If $\lim_{x \rightarrow \infty} f(x) = \pm\infty$, then $\lim_{n \rightarrow \infty} a_n = \pm\infty$.

Remark. The converse is not true. E.g., if $f(x) = \sin(\pi x)$, then

$$a_n = f(n) = 0 \rightarrow 0 \text{ as } n \rightarrow \infty,$$

but $\lim_{x \rightarrow \infty} f(x)$ does not exist.

Due to the Theorem, many properties on limits of functions can be transferred to limits of sequences.

Properties of Convergent Sequences. Suppose $\{a_n\}$ and $\{b_n\}$ are convergent and $c \in \mathbb{R}$.

- 1) $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$.
- 2) $\lim_{n \rightarrow \infty} ca_n = c \lim_{n \rightarrow \infty} a_n$.
- 3) $\lim_{n \rightarrow \infty} a_n b_n = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$.
- 4) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$ (provided that $\lim_{n \rightarrow \infty} b_n \neq 0$).
- 5) $\lim_{n \rightarrow \infty} (a_n)^p = \left(\lim_{n \rightarrow \infty} a_n \right)^p$ if $p > 0$ and $a_n \geq 0$.

The Squeeze Theorem. Suppose there is n_0 such that $a_n \leq b_n \leq c_n$ for all $n \geq n_0$.

If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Since $-|a_n| \leq a_n \leq |a_n|$, we get the following special case of the Squeeze Theorem.

Fact 1. If $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$.

Remark. $\lim_{n \rightarrow \infty} |a_n| = L \neq 0 \not\Rightarrow \lim_{n \rightarrow \infty} a_n$ exists.

E.g., if $a_n = (-1)^n$, then $\lim_{n \rightarrow \infty} |a_n| = 1$ but $\lim_{n \rightarrow \infty} a_n$ does not exist.

Fact 2. If $a_n \rightarrow L$, then every subsequence of $\{a_n\}$ is also convergent to L .

Fact 3. A sequence $\{a_n\}$ is divergent if it has two subsequences $\{a_{n_j}\}_{j=1}^{\infty}$ and $\{a_{m_k}\}_{k=1}^{\infty}$ such that $\lim_{j \rightarrow \infty} a_{n_j} = L$, $\lim_{k \rightarrow \infty} a_{m_k} = M$ and $L \neq M$.

Example 1. Determine the convergence or divergence of the following sequences.

(i) $a_n = \frac{n}{2n+1}$:

We have $\lim_{n \rightarrow \infty} \frac{n}{2n+1} = \frac{1}{2}$, since $\lim_{x \rightarrow \infty} \frac{x}{2x+1} = \lim_{x \rightarrow \infty} \frac{1}{2 + \frac{1}{x}} = \frac{1}{2}$.

(ii) $a_n = \frac{\ln n}{n}$:

We have $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$, since $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$ by L'Hospital's Rule.

(iii) $a_n = \frac{(\cos n)^2}{n}$:

Since $0 \leq (\cos n)^2 \leq 1$, we have $0 \leq a_n \leq \frac{1}{n}$. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} a_n = 0$.

(iv) $a_n = \frac{n!}{n^n}$:

Now $0 < a_n = \frac{1 \cdot 2 \cdots n}{n \cdot n \cdots n} \leq \frac{1}{n}$. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} a_n = 0$.

(v) $a_n = 2^{(-1)^n}$:

Now $a_{2n-1} = \frac{1}{2} \rightarrow \frac{1}{2}$ and $a_{2n} = 2 \rightarrow 2$. By Fact 3, $\{a_n\}$ is divergent.

(vi) $a_n = 1 + \frac{(-1)^n}{n}$:

Since $\left| \frac{(-1)^n}{n} \right| = \frac{1}{n} \rightarrow 0$, $\frac{(-1)^n}{n} \rightarrow 0$. Hence, $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 1$. ■

One of the properties on limits of sequences mentioned earlier is that if $p > 0$ and $a_n \geq 0$, then $\lim_{n \rightarrow \infty} (a_n)^p = \left(\lim_{n \rightarrow \infty} a_n \right)^p$. This is true because the function $y = x^p$ is continuous.

More general, we have

If $\lim_{n \rightarrow \infty} a_n = L$ and $g(x)$ is a continuous function, then $\lim_{n \rightarrow \infty} g(a_n) = g(L)$.
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In particular, we have

Fact 4. If $\lim_{n \rightarrow \infty} a_n = L$, then $\lim_{n \rightarrow \infty} |a_n| = |L|$.

Example 2. Determine the convergence or divergence of the following sequences.

(i) $a_n = e^{1/n^2}$: Since $\frac{1}{n^2} \rightarrow 0$ and $g(x) = e^x$ is continuous, we have $\lim_{n \rightarrow \infty} a_n = e^0 = 1$.

(ii) $a_n = \ln(n+1) - \ln n$: $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n} \right) = \ln 1 = 0$, since $g(x) = \ln x$ is continuous.

(iii) $a_n = \cos \left(\frac{\pi}{n} \right)$: $\lim_{n \rightarrow \infty} a_n = \cos 0 = 1$, since $g(x) = \cos x$ is continuous.

(iv) $a_n = r^n$: Recall that

$$\lim_{x \rightarrow \infty} r^x = \begin{cases} \infty & \text{if } r > 1 \\ 1 & \text{if } r = 1 \\ 0 & \text{if } 0 < r < 1 \end{cases}.$$

Thus we know $\lim_{n \rightarrow \infty} r^n$ for $r > 0$. Now suppose $r \leq 0$. Then

when $-1 < r \leq 0$, $|r^n| = |r|^n \rightarrow 0$ and hence $r^n \rightarrow 0$ (by Fact 1);

when $r = -1$, $\{r^n\} = \{(-1)^n\}$ is divergent (by Fact 3);

when $r < -1$, $|r^n| = |r|^n \rightarrow \infty$ and hence $\{r^n\}$ is divergent (by Fact 4).

We conclude that the sequence $\{r^n\}$ is convergent exactly when $-1 < r \leq 1$. ■

Two Concepts on Sequences.

(i) A sequence $\{a_n\}$ is called monotonic if $\{a_n\}$ is either increasing or decreasing. That is, we have either $a_1 \leq a_2 \leq a_3 \leq \cdots$ or $a_1 \geq a_2 \geq a_3 \geq \cdots$.

If $\{a_n\}$ is increasing, then a_1 is a lower bound for $\{a_n\}$; if $\{a_n\}$ is decreasing, then a_1 is an upper bound for $\{a_n\}$.

(ii) A sequence $\{a_n\}$ is called bounded above if there exists $M > 0$ such that $a_n \leq M$ for all n ; $\{a_n\}$ is called bounded below if there exists $m > 0$ such that $a_n \geq m$ for all n .

A sequence $\{a_n\}$ is called bounded if it is both bounded above and bounded below.

Example 3. (1) If $a_n = \frac{3}{n+5}$, then $\{a_n\}$ is decreasing and bounded with $0 < a_n \leq \frac{1}{2}$.

(2) If $a_n = \frac{n^2}{n^2+1}$, then $\{a_n\}$ is increasing and bounded with $\frac{1}{2} \leq a_n \leq 1$.

(3) If $a_n = \frac{n!}{n^n}$, then $\{a_n\}$ is decreasing and bounded with $0 < a_n \leq 1$, since

$$a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}} = \frac{n!}{(n+1)^n} < \frac{n!}{n^n} = a_n. \quad \blacksquare$$

Monotonic Sequence Theorem. Every bounded monotonic sequence is convergent.

More precisely, we have

if $\{a_n\}$ is increasing and bounded, then $\lim_{n \rightarrow \infty} a_n =$ the least upper bound of $\{a_n\}$;

if $\{a_n\}$ is decreasing and bounded, then $\lim_{n \rightarrow \infty} a_n =$ the greatest lower bound of $\{a_n\}$.

Fact 5. If $a_n = \left(1 + \frac{1}{n}\right)^n$, then $\{a_n\}$ is increasing and bounded, and we have

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e.$$

Example 4. Let $a_n = \left(1 + \frac{1}{2n}\right)^n$ and $b_n = \left(1 - \frac{1}{n}\right)^n$. Then

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{\left(1 + \frac{1}{2n}\right)^{2n}} = \sqrt{e}$$

and

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{n-1} \left(1 - \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n-1}\right)^{n-1}} \left(1 - \frac{1}{n}\right) = \frac{1}{e}. \quad \blacksquare$$

Section 11.2: Series

A series of real numbers has the form

$$a_1 + a_2 + a_3 + \cdots, \quad \text{which is also denoted as } \sum_{n=1}^{\infty} a_n,$$

where $\{a_n\}$ is an infinite sequence of real numbers.

But what does it mean? We know very well the meaning of the following:

$$a_1, \quad a_1 + a_2, \quad a_1 + a_2 + a_3, \quad \cdots \quad \cdots.$$

Let us write

$$s_1 = a_1, \quad s_2 = a_1 + a_2, \quad \cdots, \quad s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k.$$

Then $\{s_n\}$ is a sequence of real numbers, called the sequence of the partial sums of $\sum_{n=1}^{\infty} a_n$.

Definition. Let $\sum_{n=1}^{\infty} a_n$ be a series and let $\{s_n\}$ be the sequence of partial sums of $\sum_{n=1}^{\infty} a_n$.

If $\lim_{n \rightarrow \infty} s_n = s$ exists, then we say that the series $\sum_{n=1}^{\infty} a_n$ is convergent and we write $\sum_{n=1}^{\infty} a_n = s$, where s is called the sum of $\sum_{n=1}^{\infty} a_n$.

If $\{s_n\}$ is not convergent, then we say that the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Remark. When $\lim_{n \rightarrow \infty} s_n = \pm\infty$, we also write $\sum_{n=1}^{\infty} a_n = \pm\infty$.

Example 1. A geometric series has the form

$$a + ar + ar^2 + ar^3 + \cdots = \sum_{n=1}^{\infty} ar^{n-1},$$

where a and r are constants and $a \neq 0$. Now we see that

$$s_n = a + ar + \cdots + ar^{n-1} = \begin{cases} \frac{a(1-r^n)}{1-r} & \text{if } r \neq 1 \\ na & \text{if } r = 1 \end{cases}.$$

Thus $\lim_{n \rightarrow \infty} s_n$ exists exactly when $|r| < 1$; in this case, $\lim_{n \rightarrow \infty} s_n = \frac{a}{1-r}$. Therefore,

$$\sum_{n=1}^{\infty} ar^{n-1} = \begin{cases} \frac{a}{1-r} & \text{if } -1 < r < 1 \\ \infty & \text{if } r \geq 1 \text{ and } a > 0 \\ -\infty & \text{if } r \geq 1 \text{ and } a < 0 \\ \text{divergent} & \text{if } r \leq -1 \end{cases}. \quad \blacksquare$$

Example 2. Determine the convergence or divergence of the following series.

$$(1) \quad 5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \frac{80}{81} - \cdots.$$

This is a geometric series with $a = 5$ and $r = -\frac{2}{3}$. Since $|r| < 1$, the geometric series is convergent with sum $s = \frac{a}{1-r} = 3$.

$$(2) \quad \sum_{n=1}^{\infty} 2^{2n} 3^{1-n}.$$

Now the series has the form $\sum_{n=1}^{\infty} 3 \frac{4^n}{3^n} = \sum_{n=1}^{\infty} 4 \left(\frac{4}{3}\right)^{n-1}$ and hence it is a geometric series

with $a = 4$ and $r = \frac{4}{3} > 1$. Therefore, $\sum_{n=1}^{\infty} 2^{2n} 3^{1-n} = \infty$.

$$(3) \quad \sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$$

Now $a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. Hence,

$$s_n = \sum_{k=1}^n a_k = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1} \rightarrow 1 \text{ as } n \rightarrow \infty.$$

Therefore, $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$ is convergent.

$$(4) \quad \sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right).$$

Now $a_n = \ln n - \ln(n+1)$. Hence,

$$s_n = \sum_{k=1}^n a_k = \sum_{k=1}^n \left(\ln k - \ln(k+1) \right) = -\ln(n+1) \rightarrow -\infty \text{ as } n \rightarrow \infty.$$

Therefore, $\sum_{n=1}^{\infty} \ln \left(\frac{n}{n+1} \right) = -\infty$ is divergent. ■

Remark. Series in (3) and (4) are called telescoping series, which has the form $\sum_{n=1}^{\infty} (t_n - t_{n+1})$.

Example 3. Write the number $2.3\overline{17}$ as a ratio of integers.

Now we have

$$\begin{aligned} 2.3\overline{17} &= 2.3171717\ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \cdots \\ &= \frac{23}{10} + \frac{17}{10^3} + \frac{17}{10^3} \left(\frac{1}{10^2} \right) + \frac{17}{10^3} \left(\frac{1}{10^2} \right)^2 + \cdots \\ &= \frac{23}{10} + \frac{\frac{17}{10^3}}{1 - \frac{1}{10^2}} = \frac{23}{10} + \frac{17}{990} = \frac{1147}{495}. \quad \blacksquare \end{aligned}$$

Example 4. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ is divergent.

We see that

$$s_2 = 1 + \frac{1}{2},$$

$$s_4 = s_2 + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \frac{1}{2} = 1 + \frac{2}{2},$$

$$s_8 = s_4 + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{2}{2} + \frac{1}{2} = 1 + \frac{3}{2},$$

... ..

$$s_{2^n} > 1 + \frac{n}{2} \rightarrow \infty \text{ as } n \rightarrow \infty.$$

Since $\{s_n\}$ is increasing, $\lim_{n \rightarrow \infty} s_n = \infty$; that is, $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ is divergent. ■

Note that if $\sum_{n=1}^{\infty} a_n = s$, then $a_n = s_n - s_{n-1} \rightarrow s - s = 0$. Therefore, we have

Theorem. If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n \rightarrow \infty} a_n = 0$.

The Test for Divergence. If $\lim_{n \rightarrow \infty} a_n$ DNE or $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Example 5. By the Test for Divergence, $\sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{2}\right)$ and $\sum_{n=1}^{\infty} \frac{n^2}{2n^2 + 3}$ are divergent. ■

Remark. $\lim_{n \rightarrow \infty} a_n = 0 \not\Rightarrow \sum_{n=1}^{\infty} a_n$ is convergent. E.g., $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ but $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$.

Properties of Convergent Series. Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent. Then

$$(i) \quad \sum_{n=1}^{\infty} (a_n \pm b_n) = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n;$$

$$(ii) \quad \sum_{n=1}^{\infty} c a_n = c \sum_{n=1}^{\infty} a_n.$$

Example 6. $\sum_{n=1}^{\infty} \left(\frac{1}{n(n+1)} + \frac{5}{2^n} \right) = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} + 5 \sum_{n=1}^{\infty} \frac{1}{2^n} = 1 + 5 = 6. \quad \blacksquare$

Remark. $\sum_{n=1}^{\infty} a_n b_n \neq \left(\sum_{n=1}^{\infty} a_n \right) \left(\sum_{n=1}^{\infty} b_n \right)$ and $\sum_{n=1}^{\infty} \frac{a_n}{b_n} \neq \frac{\sum_{n=1}^{\infty} a_n}{\sum_{n=1}^{\infty} b_n}.$

Section 11.3: The Integral Test for Positive Series

Suppose $f : [1, \infty) \rightarrow (0, \infty)$ is continuous and decreasing. Let $a_n = f(n)$. Then the

Integral Test says $\int_1^\infty f(x) dx$ and $\sum_{n=1}^\infty a_n$ have the same convergence/divergence:

$$\int_1^\infty f(x) dx \text{ is convergent} \iff \sum_{n=1}^\infty a_n \text{ is convergent};$$

$$\int_1^\infty f(x) dx \text{ is divergent} \iff \sum_{n=1}^\infty a_n \text{ is divergent}.$$

Remark. In the above, $[1, \infty)$ can be replaced by $[a, \infty)$ and we can only require that $a_n = f(n)$ for all $n \geq n_0$.

Why we have the Integral Test. Let $s_n = \sum_{k=1}^n a_k$ and $A_n = \int_1^n f(x) dx$. Since

$$a_{k+1} \leq \int_k^{k+1} f(x) dx \leq a_k,$$

we have $s_n - s_1 \leq \int_1^n f(x) dx = A_n \leq s_{n-1}$. That is,

$$s_n \leq s_1 + A_n \leq s_{n-1} \leq s_n.$$

Hence, $\{A_n\}$ is bounded $\iff \{s_n\}$ is bounded. Therefore, the Integral Test holds. ■

Example 1. (i) The series $\sum_{n=1}^\infty \frac{1}{n^2 + 1}$ is convergent, since

$$\int_1^\infty \frac{1}{x^2 + 1} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2 + 1} dx = \lim_{t \rightarrow \infty} \left[\tan^{-1} x \right]_1^t = \lim_{t \rightarrow \infty} \tan^{-1}(t) - \frac{\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

is convergent. Note that now $f(x) = \frac{1}{x^2 + 1} > 0$ is continuous and decreasing on $[1, \infty)$.

(ii) The series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is divergent, since

$$\int_1^{\infty} \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{2} (\ln x)^2 \right]_1^t = \lim_{t \rightarrow \infty} \frac{1}{2} (\ln t)^2 = \infty$$

is divergent. Note that now $f(x) = \frac{\ln x}{x} > 0$ is continuous and decreasing on $[3, \infty)$, since $f'(x) = \frac{1 - \ln x}{x^2} < 0$ if $x \geq 3$.

Example 2. We consider the convergence/divergence of the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$.

When $p \leq 0$, $\frac{1}{n^p} \not\rightarrow 0$ and hence the series is divergent by the Test for Divergence.

For $p > 0$, let $f(x) = \frac{1}{x^p}$. Then f is continuous and decreasing on $[1, \infty)$. When

$0 < p < 1$, $\int_1^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \frac{1}{1-p} (t^{1-p} - 1) = \infty$ is divergent;

when $p = 1$, $\int_1^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} \ln t = \infty$ is divergent; when $p > 1$,

$\int_1^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \frac{1}{1-p} (t^{1-p} - 1) = \frac{1}{p-1}$ is convergent.

Therefore, the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent exactly when $p > 1$. ■

Example 3. By Example 2, we see that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}, \quad \sum_{n=1}^{\infty} \frac{1}{n^3}, \quad \sum_{n=1}^{\infty} \frac{1}{n^{1.02}}, \quad \text{etc.}$$

are convergent, and the series

$$\sum_{n=1}^{\infty} \frac{1}{n}, \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}, \quad \sum_{n=1}^{\infty} \frac{1}{n^{0.8}}, \quad \text{etc.}$$

are divergent. ■

Remark. In the next section, we will consider Comparison Test for positive series. As we know the convergence/divergence of all p -series and geometric series, they are often used to determine the convergence/divergence of other series.